

SHORTCUT BEFORE CIRCUIT: DOCUMENT STATISTICS TIME IN-CONTEXT CONFLICT RESOLUTION

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ABSTRACT

When a context asserts two different values for the same fact, a model must commit to one cue—recency, repetition count, or position. Which cue it uses is hard to establish, because in natural data the cues agree on nearly every document. We train 26M-parameter transformers from scratch on a synthetic assignment language in which a recency rule and a rarity rule are exactly coextensive on the training distribution, and separate them with a minimal causal edit that inverts one cue while holding token count, answer position, and all other tokens fixed. Every analyzed configuration reaches 100% in-distribution accuracy, yet under intervention the same models diverge, and the readout is ordered by redundancy: with a superseded value repeated 16 times it is rarity-type in 0.96–1.00 of documents, while with 3 repetitions the sign reverses in every column (0.09–0.41). A magnitude-matched control edit on a non-queried slot stays within 0.015 nats. Training first passes through a positional-shortcut phase whose accuracy ceiling we derive in closed form from the token layout and observe saturated to within 2% at narrow support, falling to 70–73% as the support widens, with the retrieval circuit absent; the step at which a model leaves this phase is set by both statistics at a fixed optimization schedule. Attribution performed during the phase reverses the sign of the conclusion, with no in-distribution accuracy change to signal it, so a phase diagram of this kind must gate on circuit formation rather than on task accuracy.

1 INTRODUCTION

A document may assert that a slot holds one value and later assert that it holds another. Answering a query about that slot requires committing to a cue, and in ordinary text the cues agree: the value stated later is usually also the value stated once rather than many times. Because they agree, accuracy reveals nothing about which cue a model uses. The commitment surfaces only when they come apart—a stale retrieval corpus, a partially edited document, an inconsistently propagated update—and by then the choice was fixed during pretraining.

Documents in our synthetic assignment language are sequences of (entity, attribute, value) statements followed by a query; the answer is the most recent value assigned to the queried slot. Bindings are resampled per document and the training stream is infinite, so the task cannot be solved from weights. We vary two statistics of the cue structure while holding the task, its Bayes-optimal policy, and the entire training recipe fixed: R_{old} , the number of times a superseded value is repeated, and ΔD , the number of statements from the last update to the query. Because the current value appears exactly once and each superseded value appears R_{old} times, “take the rarest value” and “take the most recent value” select the same value on every training document, so no observational analysis can separate them. We therefore read out the rule with a minimal causal edit that inverts the multiplicities of the two competing values, leaving the ground truth, the token count, the answer position, and every other token unchanged. The signed change in log-odds, Δ , is positive when the readout tracks rarity and negative when multiplicity boosts the value the model should suppress.

Every analyzed cell reaches $\text{acc} = 1.000$, including on the stratum where the trivial “the last update is the answer” heuristic fails. Under intervention the same models separate, and the readout is ordered by redundancy: at $R_{\text{old}} = 16$ it is rarity-type with sign fractions 0.96–1.00, while at $R_{\text{old}} = 3$ the sign reverses in every ΔD column (0.09–0.41). Two mechanisms with opposite dependence

on repetition count receive the same score from any held-in evaluation. A control edit that inverts multiplicities on a non-queried slot perturbs the same number of tokens and stays within 0.015 nats, so the effect is not an artifact of edit magnitude.

Before acquiring retrieval, models converge on a rule that copies the value at a fixed token offset from the end. Its accuracy is computable: statements occupy exactly four tokens and the answer lies $1 + \Delta D$ statements from the end, so the best fixed offset succeeds exactly when ΔD takes its modal value, giving a ceiling of $1/|\text{supp}(\Delta D)|$ that models saturate to within 2% where the support is narrow, and attain 70–73% of where it is wide. Throughout the plateau, accuracy on repeated value positions sits near chance: the retrieval circuit does not exist yet. Applying our probe during this phase yields strongly negative Δ , which read as a rule attribution would place a clean frequency-type region at low R_{old} —an artifact produced by models with no retrieval circuit at all. Within a single run Δ crosses from -0.587 to $+1.802$ across one evaluation interval, with in-distribution accuracy unchanged at both ends. We therefore gate every cell on circuit formation before reporting a mechanism, and report the gate and the resulting three-way state for every cell.

2 RELATED WORK

Behavioral studies of knowledge conflict establish that models resolve conflicting evidence systematically rather than arbitrarily. Preferences track entity popularity and evidence coherence (Longpre et al., 2021; Xie et al., 2024), the plausibility of the update (Kortukov et al., 2024), the number of agreeing sources (Chen et al., 2022; Jin et al., 2024a), surface repetition (Yan et al., 2024), and source attributes such as authority (Schuster et al., 2026). These measure which cue wins in a deployed model but cannot attribute the preference to data: the pretraining corpus is fixed and, for the cues at issue, unobservable, and because natural documents rarely make the candidate cues disagree, any observed preference is confounded with whatever cue happened to correlate with the answer.

Two lines address the mechanism directly. Interpretability work localizes the components that carry conflicting information (Jin et al., 2024b) and identifies prefix-matching induction heads as the canonical in-context retrieval circuit (Olsson et al., 2022), whose formation has been traced in detail (Singh et al., 2024) and shown to be non-monotone (Singh et al., 2023). Controlled-corpus studies show that distributional properties govern whether in-context learning appears at all—burstiness and Zipfian skew (Chan et al., 2022), task diversity (Raventós et al., 2023), the balance between look-up and in-weight solutions (Bratulić et al., 2025), corpus composition shifting parametric against in-context knowledge (Kim et al., 2025)—with corpus-side evidence that supportive documents carry rare tokens and long-range dependencies (Han et al., 2023). In both lines the contrast is whether in-context retrieval is used, or context against weights. The cues within the context are never separated, because natural data does not separate them.

Our positional rule is a shortcut in the sense of Geirhos et al. (2020); McCoy et al. (2019), but unusual in that the token layout is ours to fix, so its payoff follows in closed form and can be moved along a controlled axis rather than diagnosed after the fact. Abrupt reorganizations during training are often summarized as phase diagrams (Power et al., 2022; Nanda et al., 2023; Doshi et al., 2024; Goddard et al., 2025), and claims of abruptness require care since discontinuous metrics can manufacture transitions from smooth change (Schaeffer et al., 2023); we report the transition in unthresholded metrics alongside the gate. Park et al. (2024) is closest: a phase diagram for a synthetic in-context task, with phases framed as competition between algorithms. Their axes are properties of the task distribution and phases are assigned by matching model outputs against candidate algorithms; our axes are statistics of the cue structure within a document, with the task and its Bayes-optimal policy identical in every cell, so the in-distribution optimum is attained everywhere and behavioral matching cannot distinguish the phases. Their result shows task-level statistics selecting among algorithms for *what* to compute; ours shows document-level cue statistics selecting among mechanisms for the *same* computation, and adds a competitor whose payoff is known in closed form.

3 SETUP

3.1 A LANGUAGE WITH AN ALIASED RULE PAIR

Documents are sequences of assignment statements followed by a query. Each statement binds a value to an (entity, attribute) pair, which we call a *slot*, and is emitted as four tokens [ent] [attr] [val] [SEP]; the query appends [QUERY] [ent] [attr] [ARROW] and the target is the value currently bound to the queried slot. Identifiers come from disjoint vocabularies of size 200, 8 and 512. Bindings are resampled per document and the stream is infinite, so the task is not solvable from weights.

Every document contains one queried slot: a value v_{old} is bound to it and repeated R_{old} times, a later statement rebinds the slot to v_{new} , and the query follows ΔD statements after that rebinding. Remaining positions are filled by statements from other slots, each independently receiving an update with probability $p_{\text{update}} = 0.5$. Values are drawn without replacement within a document, so each identifier belongs to at most one slot; this makes “the answer value occurs exactly once before the query” structural rather than probabilistic, removing a baseline noise term from every measurement.

The alias. Because each superseded value is repeated R_{old} times while the current value is written once, for the queried slot s of any document d

$$\underbrace{\arg \max_i \text{pos}(s_i)}_{\text{RECENCY}} = \underbrace{\arg \min_v |\{i : \text{val}(s_i) = v\}|}_{\text{RARITY}} \quad \text{for all } d, \quad (1)$$

with ties in RARITY broken toward the more recent occurrence. Equation 1 is an identity of the generator, not a statistical tendency: the two rules never disagree on a single training document, the objective is indifferent between them, and no observational analysis can separate them. Two further candidates are near-aliases. FREQUENCY coincides with the target on 0.968–1.000 of documents, degenerating to RECENCY when realized redundancy is 1 and to PRIMACY otherwise; GLOBAL LAST UPDATE coincides whenever no update follows the queried rebinding, with probability $(1 - \rho)^{\Delta D}$ for $\rho = p_{\text{update}} / (1 + p_{\text{update}} R_{\text{old}})$, which spans 0.04–0.90 across the grid (Table ??). Both are therefore also unavailable to observational attribution over much of the grid (per-cell collision rates: App. I).

The alternative construction — making the label itself ambiguous, e.g. returning v_{old} with probability $1 - p$ — does not answer our question: the optimal solution is then to reproduce p , the objective fixes the answer, and the measurable quantity becomes calibration error rather than mechanism identity. Aliasing leaves mechanism choice underdetermined by the loss while keeping a single deterministic ground truth, so accuracy remains a meaningful convergence criterion.

3.2 THE TWO AXES

Neither axis changes the target function; both change how expensive each mechanism is to implement.

Redundancy $R_{\text{old}} \in \{3, 5, 8, 12, 16\}$, fixed per configuration, sets how often the language demonstrates that a value appearing more than once is not the current one. Two lower values are excluded. At $R_{\text{old}} = 1$ the two values occur once each, RARITY carries no information, and our edit (§4) has no domain: measured yield is 0.000 in every column. At $R_{\text{old}} = 2$ the retrieval circuit fails to form within our budget in three of five columns (§??); we report all five cells in App. H as evidence for the two-phase account rather than extending the budget for one column, which would confound R_{old} with total training. The upper end is structural: superseded copies are placed in a window of width $\text{spread} \times n_{\text{stmts}}$ before the rebinding, and once $2R_{\text{old}}$ exceeds that width the window degenerates into a solid block abutting the rebinding, making the queried slot identifiable without reading the query. At $\text{spread} = 0.8$ and $n_{\text{stmts}} \geq 45$ the largest admissible value is 16.

Update-to-query distance ΔD is the number of statements between the final rebinding and the query. A recency mechanism must match the queried slot and select its latest occurrence, attending across the ΔD intervening statements; suppressing repeated values within a slot has no such dependence. We sample $\Delta D \sim U[\ell(d), h(d)]$ with $\ell(d) = \max(1, \text{round}(d/2))$ and $h(d) = \max(\ell+1, \text{round}(3d/2))$ for nominal $d \in \{2, 3, 5, 8, 16\}$.

The support is an interval rather than a point for a specific reason. With ΔD constant the answer sits at a fixed token offset from the end of every document, and “copy the value at offset k ” becomes a fully correct rule requiring no slot matching. The width of the support bounds the accuracy of such a rule at $1/|\text{supp}(\Delta D)|$, which falls from 0.333 at $d \in \{2, 3\}$ to 0.059 at $d = 16$ (Table ??). We call this the *positional ceiling* and treat it as a first-class quantity, because models occupy it for thousands of steps before acquiring retrieval (§??). Note that $d = 2$ and $d = 3$ share a support size and hence a ceiling: they serve as an internal replicate, two independent cells whose plateau heights should agree if the closed-form account is correct.

3.3 INVARIANTS AND TRAINING

Because the argument rests on the two axes being the only systematic difference between cells, we verify five invariants on 1500 documents per cell before training: bindings are not memorizable, ΔD is realized exactly and is uncorrelated with document length, the answer never equals the last value token before the query, update positions are uniform in the pre-query region, and the queried slot is matched to filler slots on same-slot adjacency (within 0.041 in every cell) while its mean distance to the nearest same-slot antecedent exceeds the filler mean by 2–22%, a gap that follows from spreading R_{old} copies over a fixed window and therefore varies along both axes (Table ??). App. A reports measured values and describes three earlier generator versions in which one of these failed — in one case with a bias that scaled in R_{old} , collinear with our main axis and therefore invisible to any cross-cell comparison. Document length is sampled from $U[45, 55]$ statements independently of both axes, with the lower bound set by the most demanding cell ($n_{\text{stmts}} \geq R_{\text{old}} + \Delta D_{\text{max}} + 5$) and then used everywhere, since a cell-dependent length distribution would itself be a covariate. We deliberately do not equalize the tail-update fraction $(1 - \rho)^{\Delta D}$: forcing it constant requires inserting distractor updates in the tail, which at small ΔD creates a different perfect shortcut, so we report it as a covariate and stratify held-in evaluation on it.

We train decoder-only transformers with 8 layers, $d_{\text{model}} = 512$, 8 heads, rotary embeddings and pre-norm blocks (26.1M parameters). Attention uses per-head RMS normalization on queries and keys with a learned gain initialized at 2.0 rather than 1.0; App. B derives the logit bound that makes this necessary at our context lengths. Each configuration is trained for 16000 steps at batch size 256 with AdamW (lr = 10^{-3} , cosine decay, weight decay 0.1, none on embeddings) in bfloat16, with loss on all tokens and documents generated on the fly. The budget is set by the slowest cell rather than the fastest: the circuit forms before step 1000 at $R_{\text{old}} = 16$ but near step 6000 at $R_{\text{old}} = 3$ (§??), and 16000 steps leaves every analyzed cell at least 10000 post-formation steps. App. E shows the readout is stable across an $8\times$ budget range in the interior of the grid; App. A documents a dataloader failure mode in which reusing the same seed offset across passes replays a fixed document set, driving training loss below the analytic entropy floor while held-out loss rises and producing an apparent phase transition that is entirely memorization.

4 READING OUT THE RULE

4.1 THE MULTIPLICITY INVERSION

To separate aliased rules we apply minimal edits to held-out documents. An edit targets one rule, moves that rule’s prediction, and leaves the ground truth unchanged. Writing $m(x; v^*) = \log p(v^* | x) - \log p(v_{\text{truth}} | x)$, the readout is the paired difference

$$\Delta = m(x_{\text{edit}}; v^*) - m(x_{\text{base}}; v^*), \quad (2)$$

where both terms use the same contrast value v^* , chosen as the target rule’s post-edit prediction and required to differ from the ground truth on both sides. A predictor that assigns logit κ to the target rule’s prediction then yields $\Delta = 2\kappa$, and one that assigns it to the ground truth yields $\Delta = 0$ identically, whether or not the target rule coincides with the truth in the base document.

The edit separating RARITY from RECENCY inverts value multiplicity inside the queried slot: one copy of v_{old} is retained and every other copy is rewritten as a copy of v_{new} , so post-edit counts are $\{v_{\text{old}} : 1, v_{\text{new}} : R\}$, exchanged relative to the base. Under this edit RECENCY still predicts v_{new} (the last statement of the slot is untouched), FREQUENCY predicts v_{new} , PRIMACY predicts v_{old} on both sides and cancels in Equation 2, and only RARITY flips.

The sign of Δ therefore separates two families of mechanism rather than two rules. The edit raises the occurrence count of the correct answer from one to R and lowers the superseded value to one. Any mechanism that accumulates evidence with occurrence count — a context-frequency prior, an induction head aggregating over matches, a copying bias toward repeated tokens — predicts $\Delta < 0$; a mechanism that suppresses values whose repetition marks them as superseded predicts $\Delta > 0$.

The retained copy must be the earliest one. Our generator maintains a temporal invariant — within a slot, update flags are non-decreasing and occurrences of the same value are contiguous — so retaining an interior copy would interleave generations and produce a document outside the language. This has a useful consequence: v_{old} occupies the first position of the queried slot in both base and edited documents, so any position-based account of the readout, including attention degrading toward the first occurrence under distributional shift, contributes equally to both terms and cancels.

4.2 VALIDITY

Every edit preserves token count, answer position, ground-truth answer, statement count, and the distance from the final queried statement to its nearest same-slot antecedent; the last is not automatic, since an edit changing it would covary with ΔD in a way absent from training. We verify all five document-by-document and discard violations.

Three controls establish that Δ reflects a contest between the two rules rather than a response to the edit. Idealized predictors confirm the identity above: at $\kappa = 8$ the rule-follower returns +16.0 and the truth-follower $|\Delta| < 10^{-9}$ in every cell. A trained model with no mechanism returns zero: before the circuit forms, Δ is +0.000 with 0.49 of documents positive at step 2000 and -0.002 with 0.40 at step 4000 ($R_{\text{old}} = 3$, $\Delta D = 2$). This is stronger than the idealized check, because the network is trained rather than constructed — if the edit perturbed logits on its own, a model with nothing to move would not return zero. The same inversion applied to a filler slot perturbs the same number of tokens while leaving the queried slot untouched, and reads -0.015 to +0.017 across the grid — below 1% of the targeted readout wherever that readout exceeds 1 nat. In the low- R_{old} cells, where the median is 0.01–0.09 nats, the control is the same order of magnitude as the effect, so magnitude alone does not establish that the readout there is not edit noise; the evidence for a direction in those cells rests on the sign fraction over 400 documents and on the within-cell redundancy gradient of §5.4. The control’s sign also differs from the readout’s in three of those five cells, so it is not producing a spurious effect in the same direction.

The inversion is off-distribution in one respect: the answer value now occurs more than once before the query, which never happens during training. We therefore report the probability mass on $\{v^*, v_{\text{truth}}\}$ under the edit and treat readouts below 0.5 as invalid, applying the gate per document rather than per cell so that a cell mean cannot conceal documents on which the model has left the task. Measured mass is 0.98–1.00 across the analyzed grid; in one cell the gate moves the median by 1% at 4000 steps and 16% at 32000, the mechanism being a growing subpopulation rather than a longer tail: documents with $\Delta > 8$ nats rise from 2 of 400 at 4000 steps to 70 of 400 at 32000, approaching the analytic maximum 2κ , while the bulk of the distribution is unchanged (App. E). The edit requires at least two copies of v_{old} to survive into the document, giving a domain of 0.70–1.00 after all five invariants are re-checked, or 0.66–1.00 counting only surviving copies (Table ??); coverage falls with ΔD because copies placed outside the document canvas are discarded, so cells at low R_{old} and high ΔD condition on a subset with higher realized redundancy, reported per cell in §5.4.

4.3 WHAT WE REPORT

The primary dependent variable is the fraction of held-out documents on which Δ has the expected sign, over 400 documents at the final checkpoint, with an exact binomial test of that fraction. We report the median of Δ and its interquartile range alongside it, but treat the sign fraction as primary for three reasons.

Per-document Δ is heavy-tailed, so the mean is unstable in ways the sign fraction is not. In one cell measured at two budgets the mean shrinks threefold ($-0.74 \rightarrow -0.22$) while the sign fraction moves from chance to $p < 10^{-30}$; across 13 post-formation checkpoints of a single run the mean has standard deviation 0.49 nats while the sign fraction has 0.03 (App. E).

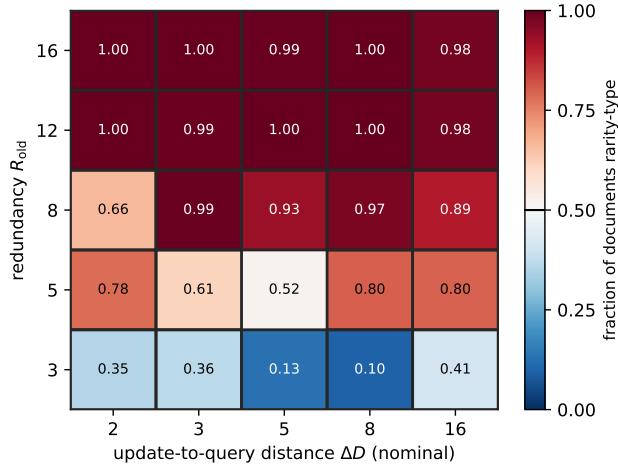


Figure 1: Causal readout across the grid at 16000 steps, seed 0. Cell entries are the fraction of 400 held-out documents on which the multiplicity inversion moves log-odds toward the rarity prediction; 0.5 is chance and the colorbar midpoint. Boxed cells are significant at $p < 0.05$ under an exact binomial test on documents (24 of 25; the exception is $R_{old} = 5, \Delta D = 5$ at $p = 0.48$). Every cell reaches in-distribution accuracy ≥ 0.999 , so no held-in evaluation separates them. Entries become means over three seeds with half-ranges.

The median is stable but bounded from above by the instrument. Contrast-pair mass falls as the effect grows, so in high-effect cells the median is compressed by however much probability has moved away from both candidates (§5.5); the sign fraction does not depend on magnitude and is unaffected. Empirically the sign fraction is monotone in R_{old} across the grid while the median is not.

The sign fraction is also the quantity that compares across seeds and budgets: it is bounded, so a saturated cell cannot drift further, and the unit of replication is the run rather than the document (§5.2).

App. J reports mean, median, interquartile range, trimmed mean and sign fraction for every cell.

5 RESULTS

5.1 BEHAVIOR SATURATES EVERYWHERE

Every analyzed cell reaches in-distribution accuracy ≥ 0.999 , and every one also reaches ≥ 0.999 on the stratum where no update follows the queried rebinding — the subset on which the trivial “the answer is the last update” heuristic fails. Held-in behavior therefore carries no signal about mechanism within the analyzed grid: any evaluation restricted to the training distribution assigns these 75 models the same score.

The premise is a claim about the analyzed grid, not about the language. Two excluded columns show what it costs. At $R_{old} = 1$ the rule pair is not aliased at all and the edit has no domain. At $R_{old} = 2$ three of five cells terminate with accuracy 0.33, 0.33 and 0.08 — values that are not near-misses but the analytic ceilings of a different mechanism (§??). At $R_{old} = 3$ all runs reach 1.000 and enter the analysis, but their readout reverses sign relative to the high end of the axis, which we report but do not interpret as a phase (§5.2). Behavioral saturation is a regime the data statistics must permit, not a given.

5.2 THE MECHANISM IS ORDERED BY REDUNDANCY

Under intervention the same models separate. Figure 1 shows the sign fraction across the grid; Table ?? gives medians, interquartile ranges and per-cell controls.

The dependence on R_{old} is monotone in the sign fraction and reverses direction at the low end. At $R_{\text{old}} = 16$ the readout is rarity-type in every column (sign fractions 0.98–1.00); it remains rarity-type at 12 (0.98–1.00) and 8 (0.66–0.99), weakens through 5 (0.52–0.80), and at $R_{\text{old}} = 3$ every column falls below chance (0.10–0.41), each significant under an exact binomial test on documents (p from 2×10^{-3} to 1×10^{-54}). Two mechanisms with opposite dependence on repetition count therefore receive the same score from any held-in evaluation.

The median tells the same story less cleanly, and the reason is instrumental rather than mechanistic. Medians run -0.09 to -0.01 at $R_{\text{old}} = 3$, 0.00 to $+0.15$ at 5, $+0.04$ to $+0.60$ at 8, and $+0.64$ to $+5.33$ at 12 — but the $R_{\text{old}} = 12$ row is non-monotone in ΔD , tracking contrast-pair mass rather than either axis (§5.5). We read the map from the sign fraction and use medians to describe effect size within a row.

The unit of replication is the run, not the document, and this bounds what the reversal supports. A per-document binomial test returns $p = 4 \times 10^{-51}$ at $R_{\text{old}} = 3$, $\Delta D = 5$, which taken alone would license a confident claim of a frequency-type mechanism; but no number of documents within one run can detect seed-level variation. We report the within-cell seed range alongside every cell (Table ??) and treat a cell-to-cell difference as meaningful only when it exceeds that range.

The ΔD dependence is weaker and we do not interpret it. At $R_{\text{old}} = 16$ the sign fraction is 0.98–1.00 in every column while the median varies by a factor of four, consistent with mass compression rather than with a mechanism gradient. Within the $R_{\text{old}} = 3$ row the sign fraction is non-monotone (0.35, 0.36, 0.13, 0.10, 0.41 at $d = 2, 3, 5, 8, 16$), a pattern we report without accounting for.

5.3 WHAT THE SIGN RULES OUT, AND WHERE

The multiplicity inversion raises the occurrence count of the correct answer from one to R and lowers the superseded value to one. Any mechanism that accumulates evidence with occurrence count predicts $\Delta < 0$: a context-frequency prior, an induction head aggregating over matches, a copying bias toward repeated tokens. We measure $\Delta > 0$ at every $R_{\text{old}} \geq 5$, with sign fraction 1.00 at $R_{\text{old}} = 16$.

At the high end of the axis the readout is therefore inconsistent with accumulation and consistent with suppression: the model down-weights values whose repetition marks them as superseded. This is the direction reported for duplicate-token and copy-suppression mechanisms in models trained on natural text, and the opposite of the majority-rule preference reported under knowledge conflict (Jin et al., 2024a; Yan et al., 2024), though neither comparison is tight — their settings supply repetition across documents rather than within a slot.

At $R_{\text{old}} = 3$ the sign reverses, which would indicate accumulation rather than suppression. We report it but do not interpret it as a phase until the seed criterion of §5.2 is applied, since the magnitude is an order of magnitude smaller than at the high end (-0.2 against $+2.4$ nats) and a reversal this small could reflect the absence of a stable mechanism rather than the presence of an opposite one. Two properties of this region are collinear and we cannot separate them — it is also where the retrieval circuit forms most slowly (§??), so seed-dependence and slow formation may be two views of the same underdetermination.

The two directions differ in shape as well as in sign. At $R_{\text{old}} = 3$ the per-document distribution is concentrated within ± 0.1 nats with a thin negative tail; at $R_{\text{old}} = 16$ it is broad, spanning 0 to 12 nats (App. J). The same edit produces near-indifference plus a few strong reactions on one side of the axis and a large shift in most documents on the other.

5.4 COVARIATES

Three quantities shift with the axes and cannot be held fixed within our construction. We report them per cell (Table ??) rather than adjusting for them.

Slot count falls monotonically in R_{old} : with length and p_{update} fixed, a document holds 27.6 distinct slots at $R_{\text{old}} = 3$ and 8.7 at 16 (Table ??). Realized redundancy is below nominal and falls with ΔD , because superseded copies placed outside the document canvas are discarded rather than compressed; compressing them would shrink the antecedent distance for the queried slot while leaving filler slots alone, which is the one invariant we most need (§3.3). Edit domain tracks realized

redundancy, from 1.00 at $R_{\text{old}} = 16$ to 0.70 at $R_{\text{old}} = 3$, $\Delta D = 16$, so the cells with the weakest readout also condition on the smallest subset.

The last of these admits a within-cell test. Realized redundancy is recorded per document, so we can split a single cell at its median and ask whether the readout tracks redundancy inside the cell as it does across cells. It does, in whichever direction that cell points. At $R_{\text{old}} = 16$, $\Delta D = 2$ the high-redundancy half reads +2.30 against +1.53 for the low half; at $R_{\text{old}} = 5$, $\Delta D = 2$, +0.083 against +0.018; and at $R_{\text{old}} = 3$, $\Delta D = 5$ — where the cell median is negative — -0.120 against -0.069, more negative rather than less. The gradient holds on both sides of zero, including in the one cell whose pooled readout is indistinguishable from zero ($R_{\text{old}} = 5$, $\Delta D = 5$: +0.013 against -0.009). Two cells at $R_{\text{old}} = 3$ show no within-cell difference ($\Delta D \in \{3, 16\}$, both within 0.002 nats).

This is the strongest evidence available to us that R_{old} acts through the statistic rather than through document selection. The comparison holds the checkpoint, the document pool, the edit and the evaluation fixed, and varies only how many copies of the superseded value a given document happens to contain. A cross-cell effect arising from edit coverage would not reproduce inside a cell where coverage is constant by construction.

5.5 WHERE THE MEASUREMENT DEGRADES

Contrast-pair mass falls as the effect grows. It is 1.00 in every $R_{\text{old}} = 3$ cell, 0.98–1.00 at $R_{\text{old}} = 5$ and 8, and as low as 0.66 at $R_{\text{old}} = 12$, $\Delta D = 5$, where the median is +5.33. This is expected — a large displacement moves probability away from both candidates — but it bounds the instrument. Gating per document at mass ≥ 0.5 shifts the cell median by 2% where mass is 0.98 and by 33% where it is 0.66 (+5.33 $\rightarrow$ +3.59), so cells whose readout exceeds roughly 5 nats cannot be separated from one another reliably.

The consequence is visible in the $R_{\text{old}} = 12$ row, where the median is non-monotone in ΔD : +3.52, +0.64, +5.33, +2.68, +1.87. The low cell is also the one with the highest mass (0.98, against 0.66 and 0.80 in its neighbours), so the ordering is consistent with how much of each effect the instrument can still register rather than with a difference in mechanism. We cannot establish that from five cells, but it is the reading the mass values support, and it is one reason we do not interpret the ΔD axis. The sign fraction is 0.98–1.00 across the entire row and is unaffected by this ceiling (§4.3).

Mass declines with R_{old} — 1.00 at 3, 0.98–1.00 at 5 and 8, 0.66–0.98 at 12, 0.82–0.98 at 16 — because the effect does. The cross-cell differences in median we report are therefore lower bounds on the underlying separation, compressed most in the cells where the effect is largest.

6 SHORTCUT DYNAMICS

Training does not proceed monotonically toward retrieval, but the route is not the same in every cell. Where the retrieval circuit forms slowly and the shortcut pays well, models spend thousands of steps on a plateau implementing a positional rule. Where it forms quickly, they pass through a low-accuracy regime and acquire retrieval without ever saturating the shortcut: at $R_{\text{old}} = 3$, $\Delta D = 2$ under a 32000-step schedule, accuracy is 0.019 at step 4000 against a ceiling of 0.333, and the copy diagnostic moves from 0.002 to 0.994 between steps 4000 and 6000. Which route a cell takes is itself a function of the axes. This section establishes three things: the plateau’s accuracy is predictable in closed form, its duration is set jointly by both document statistics at a fixed optimization schedule, and rule attribution performed on the plateau reverses the sign of the conclusion.

6.1 THE PLATEAU HAS A CLOSED-FORM CEILING AND MODELS SATURATE IT

Statements in our language occupy exactly four tokens and the queried slot’s current value lies $1 + \Delta D$ statements from the end of the sequence. A rule that copies the value at a fixed token offset from the end therefore succeeds exactly when ΔD takes the value that offset was tuned to, so the best such rule attains

$$\text{posCeil} = \max_d \Pr[\Delta D = d] = \frac{1}{|\text{supp}(\Delta D)|} \quad (3)$$

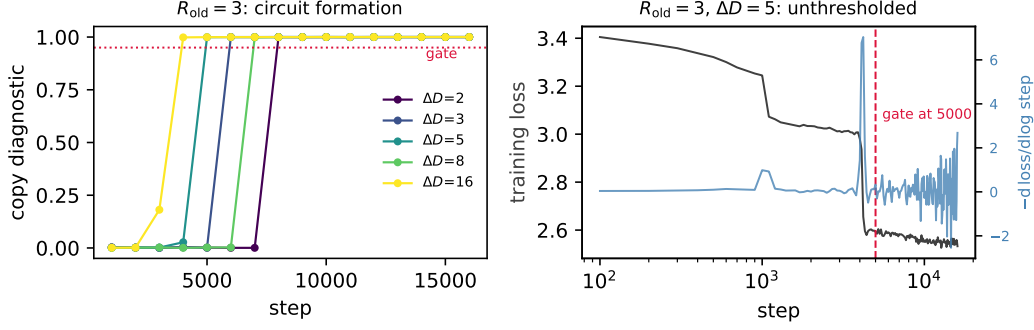


Figure 2: Left: the copy diagnostic across the $R_{\text{old}} = 3$ row. Formation is resolved only to the 1000-step evaluation grid, and the gate at 0.95 is crossed between consecutive evaluations in every column. Right: training loss for one cell, recorded every 100 steps, with its derivative against log step on the secondary axis. The unthresholded trace is shown because a discrete gate on a coarse grid can manufacture the appearance of a transition (Schaeffer et al., 2023).

the second equality holding because ΔD is uniform on its support by construction (derivation and the length-invariance argument it relies on: App. C). Table ?? compares the prediction to plateau accuracy in five cells. The two cells at $|\text{supp}| = 3$ saturate the bound to within 2%, and agree with each other to 0.0002 despite different bands — the internal replicate of §3.2. The three wider-support cells attain 70–73% of their ceilings or less. Occupancy therefore falls with support width while the bound itself falls, and we report the ratio rather than the raw accuracy for this reason. Two further cells at $R_{\text{old}} = 2$ deviate and their plateau accuracy is non-monotone in band width, which we report and analyze in App. C; those cells never escape within our budget and are excluded from the main grid.

Saturating Eq. 3 is not evidence that retrieval is absent, only that a positional rule suffices. For the stronger claim we use the copy diagnostic: accuracy on in-context value tokens that repeat an earlier value of the same slot, which requires the slot-matching computation the answer also needs but is scored on tokens the positional rule cannot reach. Throughout the plateau this diagnostic sits near chance while sequence accuracy sits at Eq. 3. A second, independent diagnostic agrees with it in every run we checked (App. D).

6.2 ESCAPE TIME IS SET JOINTLY BY BOTH STATISTICS

Escape is abrupt on the evaluation grid: the copy diagnostic moves from near chance to above 0.95 between consecutive evaluations in every cell that escapes (Figure 2, left), and sequence accuracy jumps from Eq. 3 to 1.000. (Figure 2, left). Because a discrete gate on a coarse grid can manufacture the appearance of a transition from smooth underlying change (Schaeffer et al., 2023), we also examine the training loss, which is recorded every 100 steps and involves no threshold. Its derivative with respect to log step peaks at 7.04 at step 4200 in the $R_{\text{old}} = 3$, $\Delta D = 5$ cell, against maxima of 0.13, 0.99, 1.04 and 2.08 in the step ranges $[0, 10^3)$, $[10^3, 4 \times 10^3)$, $[6 \times 10^3, 10^4)$ and $[10^4, 1.6 \times 10^4)$ respectively — a factor of seven above anything elsewhere in training, 800 steps before the gate is crossed. The last of those figures corresponds to the final decay of the cosine schedule and is not a second transition. The reorganization is therefore visible in an unthresholded, continuously recorded quantity, and the gate marks it rather than creating it.

Figure 2 (left) shows the diagnostic across the $R_{\text{old}} = 3$ row; escape steps for the full grid are tabulated in App. D. Neither axis predicts escape alone. At $R_{\text{old}} = 16$ every cell has escaped by the first evaluation regardless of band width; at $R_{\text{old}} = 3$ escape spans 4k–8k steps, broadly inverse with band width but not monotone in it; at $R_{\text{old}} = 2$ the narrow-band cell does not escape within 12k steps while the widest-band cell escapes at 5k. Because we evaluate every 1k steps, escape times are resolved only to that grid, and the $R_{\text{old}} \geq 8$ rows should be read as $\leq 1\text{k}$ rather than as measured values; distinguishing within those rows requires a finer evaluation schedule that we did not run.

The pattern is consistent with a race between two solutions. R_{old} sets how many same-slot repetitions each document supplies, and hence the supervision density available to the slot-matching computation; band width sets posCeil , and hence what the shortcut is worth to hold onto. Escape occurs when the circuit’s loss reduction overtakes the shortcut’s. We arrived at this reading only after two single-axis explanations failed—supervision density alone is contradicted by the non-monotone $R_{\text{old}} = 2$ row, and shortcut payoff alone by the $R_{\text{old}} = 3$ escapes—so we state it as the reading the grid supports, not as a mechanism we have isolated.

One qualification bounds what the axes explain. Escape time is a property of the data statistics at a fixed optimization schedule. The same cell ($R_{\text{old}} = 3$, $\Delta D = 2$), with an identical seed and data stream but a 32000-step cosine schedule, escapes at step 6000 rather than ≤ 2500 ; the two runs differ only in the learning rate at each step, by 28% at step 4000. Within a schedule the axes order the cells; across schedules the same cell moves by a comparable factor. The ordering we report is therefore relative, and the escape step should not be read as an absolute property of a corpus.

6.3 ATTRIBUTION ON THE PLATEAU REVERSES THE CONCLUSION

Applying the probe of §4 before escape yields strongly negative Δ in every cell that occupies the plateau (-0.47 with 0.05 of documents positive and -1.30 with 0.02 at $R_{\text{old}} = 2$, $\Delta D \in \{2, 8\}$; -0.94 with 0.03 in a pilot cell under an earlier vocabulary setting, App. H). Read as a rule attribution this is a clean “frequency-type” region at low R_{old} : multiplicity boosting the superseded value, exactly the pattern the majority-rule literature would predict. It is an artifact. The models producing it have no retrieval circuit, and the sign is a property of where our edit places tokens relative to the offset the positional rule reads. We have a candidate account of the mechanism but have not tested it (App. H); the claim we make here is only that the reading is not a rule attribution.

Within a single run Δ crosses zero within one evaluation interval: -0.587 with 0.08 of documents positive at step 5679, and $+1.802$ at step 6000 ($R_{\text{old}} = 3$, $\Delta D = 2$, 32000-step schedule). In-distribution accuracy is 1.000 at both points and does not change across the transition. Nothing in task performance signals that the quantity being measured has changed meaning. We therefore gate every cell on the copy diagnostic before reporting Δ , classify each run as retrieval, positional, or neither, and report the classification alongside the read-out (App. D). Read-outs in this paper are taken at a fixed budget of 16k steps; post-escape trajectories drift in both directions and are reported in App. E.

7 LIMITATIONS

Escape time depends on the optimization schedule, not only on the data. Retraining $R_{\text{old}} = 3$, $\Delta D = 2$ with an identical seed and data stream but a 32000-step cosine schedule moves circuit formation from step ≤ 2500 to 6000 and the terminal sign fraction from 0.41 to 0.89; the runs differ only in the learning rate at each step. The interior is unaffected: at $R_{\text{old}} = 16$, $\Delta D = 2$ the median is $+2.386$ at 4000 steps and $+2.433$ at 32000, with sign fractions 0.995 and 1.000. We therefore report the map at a fixed budget and treat the location of the boundary, not the existence of the phases, as budget-dependent; establishing the asymptotic boundary would require the full grid at several budgets.

The positional ceiling is collinear with ΔD . By construction $1/|\text{supp}(\Delta D)|$ falls from 0.333 to 0.059 across the axis, and the readout at $R_{\text{old}} = 16$ falls in the same direction. A fixed-width band ($\Delta D \sim U[d, d+8]$, ceiling constant at $1/9$) would break the collinearity; we report this as a planned control rather than a completed one.

Slot count is collinear with R_{old} . With length and p_{update} fixed, the number of distinct slots per document falls monotonically in R_{old} , and holding it constant would make length depend on R_{old} — two free parameters cannot satisfy three constraints. The sign reversal we report along that axis should be read as a property of the pair.

Two of our five data knobs admit no causal readout. With explicit update markers the edit changes sequence length; with history-indexed queries the target rule is no longer the last value, so Equation 2 does not apply. For these we can report observational agreement but not a mechanism, and we exclude them rather than report a readout we cannot validate.

The queried slot is not perfectly indistinguishable. Spreading R_{old} copies over a window of width $\text{spread} \times n_{\text{stmts}}$ makes the mean gap between consecutive copies 13.3 statements at $R_{\text{old}} = 3$, $\Delta D = 2$ against 10.9 for filler slots — a 22% excess that falls to 4% at $\Delta D = 16$ and sits at 11–14% across the $R_{\text{old}} = 16$ row. A model could in principle locate the queried slot from this dispersion without reading the query. Every analyzed run reaches 1.000 on the copy diagnostic, which requires genuine slot matching, so the cue is at most a redundant path; but it is collinear with both axes and we cannot rule out a contribution to the readout.

One model size, one architecture, one task family. All results are at 26.1M parameters and 8 layers on a single synthetic language. Depth plausibly shifts the boundary rather than rescaling the readout, since it controls how readily an induction-style circuit can form; we did not run the depth sweep.

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A GENERATOR INVARIANTS AND FAILURE MODES

Measured on 1500 documents per cell before training. These are properties of the generator, not of any trained model.

R_{old}	ΔD	bindMax	r_{len}	ansLast	updDens	adj_q	adj_f	ant_q	ant_f
3	2	3	-0.024	0	1.029	0.056	0.078	13.32	10.92
3	3	3	-0.023	0	1.020	0.058	0.077	13.21	10.95
3	5	3	+0.028	0	1.022	0.056	0.074	13.09	11.00
3	8	3	-0.035	0	1.021	0.060	0.077	12.55	10.92
3	16	3	-0.002	0	0.984	0.074	0.076	11.49	11.03
5	2	3	-0.003	0	1.074	0.087	0.104	9.57	8.29
5	3	3	-0.003	0	1.071	0.088	0.105	9.53	8.31
5	5	3	+0.009	0	1.013	0.090	0.107	9.30	8.34
5	8	3	+0.007	0	1.015	0.091	0.106	9.22	8.35
5	16	4	-0.013	0	1.008	0.092	0.106	8.54	8.36
8	2	3	+0.031	0	1.041	0.135	0.152	6.93	6.30
8	3	3	+0.031	0	1.031	0.135	0.151	6.91	6.32
8	5	3	-0.039	0	0.994	0.135	0.150	6.88	6.34
8	8	3	-0.003	0	0.981	0.132	0.150	6.81	6.29
8	16	3	-0.008	0	1.006	0.138	0.155	6.52	6.22
12	2	3	+0.007	0	1.102	0.180	0.197	5.35	5.00
12	3	3	+0.007	0	1.101	0.181	0.196	5.33	5.01
12	5	3	-0.036	0	1.083	0.183	0.198	5.36	5.00
12	8	3	+0.006	0	0.946	0.189	0.200	5.28	4.92
12	16	3	-0.043	0	1.056	0.188	0.206	5.16	4.84
16	2	3	+0.022	0	1.065	0.213	0.236	4.71	4.22
16	3	3	+0.022	0	1.059	0.213	0.237	4.71	4.23
16	5	2	+0.001	0	1.097	0.212	0.249	4.74	4.16
16	8	3	-0.018	0	1.052	0.213	0.248	4.69	4.16
16	16	2	-0.024	0	1.048	0.218	0.260	4.54	4.02

bindMax: the largest number of distinct documents in which any single (slot, value) triple appears. Bounded by 4 throughout, consistent with chance collision at $200 \times 8 \times 512$ possible triples and 75000 statements sampled, so bindings are not memorizable across documents. Note that counting raw occurrences rather than documents would return $\geq R_{\text{old}}$ by construction, since the queried slot holds R_{old} copies of one value within a single document.

r_{len} : Pearson correlation between realized ΔD and document token count, $|r| < 0.043$ in every cell. Without this the ΔD axis would confound distance with length.

ansLast: documents in which the answer value equals the last value token before the query, which would admit a trivial copy rule. Zero in all 25 cells, by construction rather than by chance — values are drawn without replacement within a document.

updDens: density of update statements in the final decile of the pre-query region, divided by the global density, computed over filler slots only. Within $[0.946, 1.102]$ throughout. The queried slot’s own rebinding is excluded because it sits at a fixed offset $m - \Delta D - 1$ and would land in the final decile at small ΔD , reporting the construction as a violation.

adj, ant: same-slot adjacency rate and mean distance to nearest same-slot antecedent, for the queried slot (q) and filler slots (f). Adjacency is matched within 0.041 in every cell. Antecedent distance is not: ant_q exceeds ant_f by 2–22%, largest at $R_{\text{old}} = 3$, $\Delta D = 2$ where $13.32/10.92 = 1.22$. This follows from spreading R_{old} copies over a window of width $\text{spread} \times n_{\text{stmts}} = 0.8 \times 50 = 40$ statements: $40/3 = 13.3$, matching the measured value. The gap varies along both axes and is discussed as a limitation in §7.

B QK NORMALIZATION AND THE LOGIT BOUND

Our attention applies RMS normalization to queries and keys per head before the dot product, with a learned scalar gain γ initialized at 2.0. The initialization is not cosmetic: at $\gamma = 1$ the maximum attainable attention logit is too small to express confident retrieval at our context length, and the model must reach a near-degenerate configuration to compensate.

Write $\hat{q}, \hat{k} \in \mathbb{R}^{d_h}$ for the normalized query and key, so each has unit root-mean-square per component and hence $\|\hat{q}\| = \|\hat{k}\| = \sqrt{d_h}$. The logit is

$$z = \frac{\gamma^2 \hat{q} \cdot \hat{k}}{\sqrt{d_h}}, \quad |z| \leq \frac{\gamma^2 d_h}{\sqrt{d_h}} = \gamma^2 \sqrt{d_h}, \quad (4)$$

with equality when the two vectors are exactly aligned. At $d_h = 64$ this gives $|z| \leq 8\gamma^2$. Unnormalized attention has no such bound — the logit grows with $\|q\|\|k\|$, which is free to increase during training — so the bound is a consequence of normalizing, not of our particular choice of gain.

The requirement comes from the sequence length. A document of m statements is $4m + 4$ tokens, up to 225 at $m = 55$. For a head to place fraction p of its attention on one position out of n , the target logit must exceed the others by

$$\Delta z \geq \log \frac{p(n-1)}{1-p}. \quad (5)$$

At $n = 225$ and $p = 0.9$ this is $\Delta z \geq 7.6$. With $\gamma = 1$ the available range is $[-8, +8]$, so the head can reach $p = 0.9$ only by driving $\hat{q} \cdot \hat{k}$ to within 5% of its maximum for the target position and to near zero for all 224 others. Alignment that tight is a narrow region of parameter space, and it must be found while the same head also needs intermediate confidences on other inputs. At $\gamma = 2$ the range is $[-32, +32]$ and the same $p = 0.9$ needs a normalized inner product of only 0.24 — attainable by partial alignment, with the remaining capacity available for graded preferences.

The interaction with our task is specific. The retrieval mechanism must attend from the query to one statement among $m \approx 50$ candidates, and it must do so *sharply*: the copy diagnostic of §?? scores exactly the tokens where a diffuse match fails. The positional shortcut has no such requirement — a fixed offset needs one strong position and is indifferent to the rest. Constraining logit range therefore penalizes the circuit and not its competitor, which is a confound on the very axis we measure. We set $\gamma = 2$ so that Equation 5 is satisfied with margin at every $n \leq 225$, and note that γ remains learnable, so the initialization sets where optimization starts rather than what it can reach.

Two things we did not do. We did not sweep γ ; the value follows from Equation 5 rather than from tuning. We also did not run the unnormalized variant to completion at these lengths, so we cannot report how much of the observed behavior depends on normalization as opposed to on the gain. Both are natural robustness arms and neither is in this paper.

C THE POSITIONAL CEILING

C.1 DERIVATION

Each statement occupies exactly four tokens, so statement j (zero-indexed) spans token positions $4j$ through $4j + 3$ with its value token at $4j + 2$. The query appends four tokens, giving a document of $4m + 4$ tokens for m statements.

If the query follows ΔD statements after the rebinding of the queried slot, that rebinding is statement $m - \Delta D - 1$ and its value token sits at position $4(m - \Delta D - 1) + 2 = 4m - 4\Delta D - 2$. Counting from the end of the sequence,

$$\text{offset of the answer value from the last token} = (4m + 4) - (4m - 4\Delta D - 2) = 4\Delta D + 6. \quad (6)$$

Equation 6 does not contain m . The answer’s position measured backward from the end depends only on ΔD ; document length cancels.

Consider a rule that ignores the query tokens and emits the value token at a fixed offset k from the end. By Equation 6 it is correct exactly when $4\Delta D + 6 = k$, that is when ΔD takes one particular value. Maximizing over k ,

$$\text{posCeil} = \max_d \Pr[\Delta D = d] = \frac{1}{|\text{supp}(\Delta D)|}, \quad (7)$$

the second equality holding because ΔD is uniform on its support.

C.2 WHAT THE BOUND REQUIRES

Equation 7 is an upper bound over positional rules only if ΔD cannot be predicted from anything the model observes. A rule conditioning on an observable feature correlated with ΔD could exceed it.

Two features are candidates. Document length is uninformative by Equation 6: the offset is length-independent, so length carries no signal about where the answer lies. Empirically the correlation between realized ΔD and document length is below 0.043 in every cell (Table ??). Local token content is uninformative because the queried slot’s statements are drawn from the same vocabularies and emitted in the same format as filler statements, and the unmarked language contains no update token to distinguish a rebinding from an initial binding.

One constraint deserves note. Length must satisfy $n_{\text{stmts}} \geq R_{\text{old}} + \Delta D_{\text{max}} + 5$, which at the widest cell ($R_{\text{old}} = 16$, $\Delta D_{\text{max}} = 24$) requires $n_{\text{stmts}} \geq 45$ — exactly the lower end of $U[45, 55]$. The constraint is satisfied without truncating the ΔD distribution, but it binds at the boundary: widening either axis further would induce a correlation between ΔD and length and thereby break the bound rather than merely loosen it, since a rule could then read length to predict ΔD .

The queried slot is not, however, perfectly unmarked: its copies are more dispersed than a filler slot’s by 2–22% (App. A). A rule that located the slot from this dispersion and then read its final value would exceed Equation 3, but it would also require the slot-matching computation that the copy diagnostic detects, so it is not a positional rule in the sense bounded here.

C.3 OBSERVED PLATEAU HEIGHTS

D PER-CELL STATE CLASSIFICATION

E BUDGET SENSITIVITY AND POST-FORMATION DRIFT

Every readout in this paper is taken at a fixed budget, so it matters how much of the map is a property of that choice. We ran two cells at three budgets each and tracked one of them across 13 post-formation checkpoints.

The interior is stable; the boundary is not. At $R_{\text{old}} = 16$, $\Delta D = 2$ — deep in the rarity-type region — the median moves by 2% across an $8\times$ budget range while the sign fraction is effectively saturated at both ends. At $R_{\text{old}} = 3$, $\Delta D = 2$ — on the boundary, where §5.2 declines to assign a phase — the sign fraction moves by 0.48 over the same kind of range.

cell	budget	escape	median	sign fraction
$R_{\text{old}} = 16, \Delta D = 2$	4000	1000	+2.386	0.995
	16000	1000		
	32000	2000	+2.433	1.000
$R_{\text{old}} = 3, \Delta D = 2$	12000	2508	−0.000	0.43
	16000	8000	−0.135	0.41
	32000	6000	+1.484	0.89

The two rows differ in what changes. In the first cell escape happens at the first or second evaluation under every budget, so all three readouts are taken long after the circuit exists and they agree. In the second the escape step itself moves with the schedule (§7), so the three readouts are taken at different distances past formation and they do not agree. This is the basis for reporting the location of the boundary as budget-dependent while treating the existence of the two regions as not.

Post-formation trajectories drift in magnitude but not in sign. Across the 13 evaluations from step 8000 to 32000 in the $R_{\text{old}} = 3$, $\Delta D = 2$ run at the 32000-step budget, the mean of Δ has standard deviation 0.49 nats over a range $[+1.33, +3.15]$, while the sign fraction has standard deviation 0.03 over $[0.85, 0.95]$. Accuracy is 1.000 and the copy diagnostic is 1.000 at every one of

those points, so nothing in behavior distinguishes them. This is the second of the two instabilities that motivate reporting the sign fraction rather than the mean (§4.3); the third is below.

The mass gate moves more at long budgets, and why. Gating per document on contrast-pair mass ≥ 0.5 shifts the cell median by 1% at 4000 steps and 16% at 32000 in the $R_{\text{old}} = 16$, $\Delta D = 2$ cell. The cause is visible in the per-document distribution rather than in any summary. At 4000 steps, 2 of 400 documents have $\Delta > 8$ nats and the largest is 10.86. At 32000, roughly 70 exceed 8 nats and the distribution between 8 and 14 is close to flat rather than decaying, with a maximum of 14.26 against the analytic ceiling $2\kappa = 16$. The bulk of the distribution is essentially unchanged — hence the 2% median shift — so what the longer budget produces is a subpopulation of documents on which the model has become nearly saturated, not a uniform sharpening. Because mass falls as $|\Delta|$ grows (§5.5), those are exactly the documents the gate removes.

A cross-budget consistency check on the first round. An earlier round of the grid at 12000 steps overlaps the main grid in six cells. Sign fractions agree to within 0.02 in five of them ($R_{\text{old}} = 16$ at $\Delta D \in \{2, 5, 16\}$: 1.00/1.00, 1.00/1.00, 0.98/0.96; $R_{\text{old}} = 3$, $\Delta D = 2$: 0.43/0.41) and differ by 0.20 in the sixth ($R_{\text{old}} = 5$, $\Delta D = 5$: 0.69/0.49). The pattern matches the one above — interior cells reproduce across budgets, cells near the boundary move — and it is obtained at no additional cost, since those runs exist for other reasons. The two rounds differ in step count only; we do not pool them.

F COVARIATES

Measured on 1500 documents per cell. *slots*: distinct (entity, attribute) pairs per document. R_{real} : surviving copies of v_{old} in the queried slot, below nominal because copies placed outside the document canvas are discarded. *domain*: fraction of documents with at least two surviving copies, the upper bound on edit coverage before invariant re-checking. *tail-0*: fraction with no update after the queried rebinding, matching $(1 - \rho)^{\Delta D}$ to within 1.5 points in every cell.

R_{old}	ΔD	support	posCeil	slots	R_{real}	domain	tail-0
3	2	[1, 3]	0.333	27.6	2.52	0.92	0.66
3	3	[2, 4]	0.333	27.6	2.47	0.91	0.52
3	5	[2, 8]	0.143	27.6	2.39	0.89	0.37
3	8	[4, 12]	0.111	27.6	2.23	0.81	0.19
3	16	[8, 24]	0.059	27.9	1.91	0.66	0.04
5	2	[1, 3]	0.333	20.2	3.99	0.99	0.71
5	3	[2, 4]	0.333	20.3	3.89	0.99	0.61
5	5	[2, 8]	0.143	20.4	3.72	0.97	0.48
5	8	[4, 12]	0.111	20.5	3.48	0.96	0.31
5	16	[8, 24]	0.059	20.6	2.87	0.87	0.10
8	2	[1, 3]	0.333	14.6	5.82	1.00	0.82
8	3	[2, 4]	0.333	14.6	5.71	1.00	0.73
8	5	[2, 8]	0.143	14.8	5.44	0.99	0.62
8	8	[4, 12]	0.111	14.8	5.08	0.99	0.46
8	16	[8, 24]	0.059	15.0	4.06	0.95	0.19
12	2	[1, 3]	0.333	10.8	7.84	1.00	0.86
12	3	[2, 4]	0.333	10.8	7.69	1.00	0.79
12	5	[2, 8]	0.143	11.0	7.25	1.00	0.70
12	8	[4, 12]	0.111	11.0	6.79	0.99	0.59
12	16	[8, 24]	0.059	11.1	5.42	0.98	0.34
16	2	[1, 3]	0.333	8.7	9.05	1.00	0.90
16	3	[2, 4]	0.333	8.7	8.87	1.00	0.85
16	5	[2, 8]	0.143	8.8	8.37	1.00	0.75
16	8	[4, 12]	0.111	8.9	7.81	1.00	0.65
16	16	[8, 24]	0.059	9.1	6.34	0.98	0.41

G ORACLE VALIDATION

H THE $R_{\text{old}} = 2$ CELLS

These five cells are excluded from the main grid because three of them never acquire the retrieval circuit. We report them because they are the clearest evidence for the two-phase account of §??: models that terminate on the positional shortcut, and a probe reading taken on those models that would be misread as a rule attribution.

Terminal states. All five were trained for 12000 steps under the same recipe as the main grid.

ΔD	support	posCeil	acc	copy	Δ / sign frac	state
2	[1, 3]	0.333	0.3255	0.002	-0.47 / 0.05	positional, 98% of ceiling
3	[2, 4]	0.333	0.3253	0.001	-0.44 / 0.01	positional, 98%
5	[2, 8]	0.143	0.0018	0.003	-0.00 / 0.22	neither, 1%
8	[4, 12]	0.111	0.0773	0.046	-1.30 / 0.02	positional, 70%
16	[8, 24]	0.059	1.0000	0.999	+0.33 / 0.72	retrieval, escaped at 5000

Two features of this table matter beyond the exclusion decision. The row is non-monotone in band width, analyzed in §C. And the three positional rows return Δ with the same sign and much higher internal consistency than any low- R_{old} cell in the main grid — 0.01 to 0.05 of documents positive, against 0.09 to 0.41 at $R_{\text{old}} = 3$.

Why the reading is not a rule attribution. Taken at face value, -0.47 , -1.30 and a third pilot measurement of -0.94 (sign fraction 0.03, at $R_{\text{old}} = 3$, $\Delta D = 5$ under an earlier vocabulary setting of 128 values, so not directly comparable) describe a strongly frequency-type region at low redundancy: multiplicity boosting the value the model should suppress, precisely the majority-rule pattern the knowledge-conflict literature reports. The models producing these numbers have copy diagnostics of 0.001 to 0.046. They have no retrieval mechanism to attribute. Whatever the edit is doing to their logits, it is not selecting between a recency rule and a rarity rule, because neither is implemented.

A candidate account, untested. The multiplicity inversion rewrites $R - 1$ copies of v_{old} as copies of v_{new} , leaving them at their original statement positions. A positional rule reads the value token at a fixed offset from the end of the sequence (§C). When that offset lands on a rewritten copy, the edit changes the token the rule reads from v_{old} to v_{new} , so the rule’s output flips toward v_{new} ; since $v^* = v_{\text{old}}$ for this edit, m falls and Δ is negative. The prediction is checkable without new training runs: the overlap rate between copy positions and the offset $4\Delta D + 6$ should predict the magnitude of Δ across these cells, and the account requires Δ to be near zero in the one cell that never occupied the rule — which it is (-0.00 at $\Delta D = 5$, sign fraction 0.22, the only positional-phase cell with a chance-level sign fraction). We have not run the overlap calculation, so we present this as consistent with the data rather than as established.

What we do with these cells. Nothing in the main results depends on them; they are excluded before any readout is reported. The reason for excluding by circuit formation rather than by accuracy is that accuracy does not separate them from the analyzed grid in the way one might expect. The $\Delta D = 16$ cell reaches 1.000 and enters no analysis only because $R_{\text{old}} = 2$ is outside the grid — had it been at $R_{\text{old}} = 3$ it would have been analyzed, and correctly so. Conversely a cell at 0.3255 accuracy is not a failed run in any ordinary sense: it has converged, to a different function. This is why the gate in §?? is on the copy diagnostic and not on loss or accuracy.

I COLLISION RATES AND PER-CELL READOUTS

J SUMMARY STATISTICS

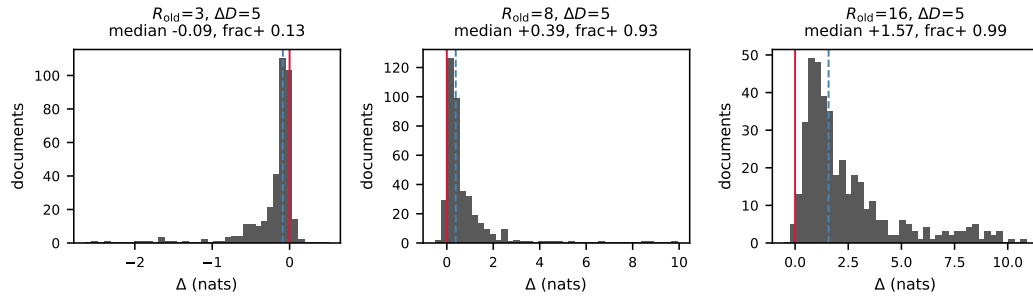


Figure 3: Per-document readout in three cells along the $\Delta D = 5$ column. Red line at zero, blue dashed at the median. Note the asymmetry in shape: the negative-side distribution is concentrated near zero with a thin tail, while the positive-side distributions are broad. Axes are not shared.